DISCRETE MATHEMATICS

Units 4 & 5

Exam revision notes, key formulas, and worked examples

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1 Unit 4 – Mathematical Logic

1.1 1. Propositions and truth values

A **proposition** is a declarative statement that is either true (T) or false (F), but not both. Questions, commands, and open expressions with unspecified variables are not propositions.

- Proposition: "Kochi is in Kerala" (T); "10 is odd" (F).
- Not a proposition: "Close the door"; "$x + 5 = 10$" without specifying x.

A truth table lists every possible assignment of truth values. For n variables, there are 2^n rows.

1.2 2. Logical connectives

Name	Symbol	True when
Negation	$\neg P$	P is false.
Conjunction (AND)	$P \wedge Q$	Both P and Q are true.
Disjunction (inclusive OR)	$P \vee Q$	At least one of P, Q is true.
Exclusive OR	$P \oplus Q$	Exactly one of P, Q is true.
Implication	$P \to Q$	False only when $P = T$ and $Q = F$.
Biconditional	$P \leftrightarrow Q$	P and Q have the same truth value.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \oplus Q$	$P \to Q$	$P \leftrightarrow Q$
T	T	F	T	T	F	T	T
T	F	F	F	T	T	F	F
F	T	T	F	T	T	T	F
F	F	T	F	F	F	T	T

Memory tip: An implication $P \to Q$ fails only in the T, F row. Also, $P \to Q \equiv \neg P \vee Q$.

1.3 3. Tautology, contradiction, contingency

- **Tautology**: always true, e.g. $P \vee \neg P$.
- **Contradiction**: always false, e.g. $P \wedge \neg P$.
- **Contingency**: true for some assignments and false for others, e.g. $P \wedge Q$.

To classify a formula, construct its truth table and inspect the final column.

1.4 4. Logical equivalences and laws

Two formulas are logically equivalent if their truth-table columns match for every assignment; write $P \equiv Q$.

- Identity: $P \wedge T \equiv P$; $P \vee F \equiv P$
- Domination: $P \vee T \equiv T$; $P \wedge F \equiv F$
- Idempotent: $P \vee P \equiv P$; $P \wedge P \equiv P$
- Double negation: $\neg\neg P \equiv P$
- Complement: $P \vee \neg P \equiv T$; $P \wedge \neg P \equiv F$
- Commutative: $P \vee Q \equiv Q \vee P$; similarly for $\wedge$
- Associative: $(P \vee Q) \vee R \equiv P \vee (Q \vee R)$; similarly for $\wedge$
- Distributive: $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$
- Distributive: $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$
- De Morgan: $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$
- De Morgan: $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$
- Absorption: $P \vee (P \wedge Q) \equiv P$
- Absorption: $P \wedge (P \vee Q) \equiv P$

$$\boxed{P \to Q \equiv \neg P \vee Q} \qquad \boxed{P \leftrightarrow Q \equiv (P \to Q) \wedge (Q \to P)}$$

1.5 5. Converse, inverse, contrapositive

For the conditional "if P, then Q" $(P \to Q)$:

Original	$P \to Q$
Converse	$Q \to P$ (swap hypothesis and conclusion)
Inverse	$\neg P \to \neg Q$ (negate both)
Contrapositive	$\neg Q \to \neg P$ (swap and negate)

The original and contrapositive are equivalent. The converse and inverse are equivalent to each other, but are not generally equivalent to the original.

1.6 6. Normal forms

- **DNF (Disjunctive Normal Form)**: OR of AND-terms, e.g. $(A \wedge B) \vee (C \wedge D)$.
- **CNF (Conjunctive Normal Form)**: AND of OR-terms, e.g. $(A \vee B) \wedge (C \vee D)$.

Useful method: eliminate $\to$ using $P \to Q \equiv \neg P \vee Q$; push negations inward using De Morgan's laws; distribute AND/OR as required by the target form.

1.7 7. Predicates and quantifiers

A **predicate** contains variable(s); its truth value depends on the variable values. Example: $P(x) : x > 5$. Assigning $x = 8$ gives a true statement; $x = 2$ gives a false one.

Symbol	Meaning	Example
$\forall x$	For every x (universal)	$\forall x\, P(x)$: every x has property P.
$\exists x$	There exists at least one x	$\exists x\, P(x)$: some x has property P.

Negating quantifiers:

$$\boxed{\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)} \qquad \boxed{\neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)}$$

Examples: "All students passed" is $\forall x\, P(x)$; "At least one student did not pass" is $\exists x\, \neg P(x)$. "No student is absent" can be written $\forall x\, \neg A(x)$, or equivalently $\neg\exists x\, A(x)$. **Nested quantifiers:** order matters in general. $\forall x \exists y\, R(x, y)$ means each x may have its own y; $\exists y \forall x\, R(x, y)$ means one fixed y works for every x.

1.8 8. Rules of inference

A valid inference uses premises to derive a conclusion.

Modus Ponens	$P \to Q,\ P\ \therefore Q$
Modus Tollens	$P \to Q,\ \neg Q\ \therefore \neg P$
Hypothetical Syllogism	$P \to Q,\ Q \to R\ \therefore P \to R$
Disjunctive Syllogism	$P \lor Q,\ \neg P\ \therefore Q$
Simplification	$P \land Q\ \therefore P$ (or Q)
Conjunction	$P,\ Q\ \therefore P \land Q$
Resolution	$P \lor Q,\ \neg P \lor R\ \therefore Q \lor R$

Exam approach: identify the premises, match their structure to a rule, and state the resulting conclusion. Do not infer P from $P \to Q$ and Q alone.

2 Unit 5 – Counting, Arrangements, and Proof

2.1 1. Addition and multiplication principles

Addition rule: use when choosing between mutually exclusive alternatives.

$$\boxed{m+n \text{ ways}}$$

Example: 5 chicken meals, 6 beef meals, and 12 vegetarian meals, with no overlap: $5+6+12=23$ choices.

Multiplication rule: use when a task has successive stages and each stage must be completed.

$$\boxed{n_1 n_2 \cdots n_k \text{ ways}}$$

Examples: coin then six-sided die: $2 \times 6 = 12$ outcomes. Five rolls of a six-sided die: $6^5 = 7776$ sequences. **Decision clue:** "either/or" often suggests addition; "first/then" or filling several positions suggests multiplication. For overlapping cases, avoid double-counting.

2.2 2. Factorials and permutations

A **permutation** is an arrangement where order matters.

$$n! = n(n-1)(n-2)\cdots 2 \cdot 1, \qquad 0! = 1$$

Arranging all n distinct objects: $n!$.

$$\boxed{{}^nP_r = \frac{n!}{(n-r)!}} \quad \text{(choose and arrange } r \text{ of } n\text{, no repetition)}$$

If repetition is allowed and each of r positions has n options: n^r. If objects repeat and identical objects are indistinguishable:

$$\boxed{\frac{n!}{p_1!p_2!\cdots p_k!}}$$

Here p_i are the frequencies of each repeated type. **Examples**

- Arrange 5 distinct books: $5! = 120$.
- Arrange 3 of 7 distinct letters without repetition: ${}^7P_3 = 7 \cdot 6 \cdot 5 = 210$.
- Six-digit PIN using 0–9, repetition allowed: $10^6 = 1,000,000$.
- ALLAHABAD has 9 letters with A repeated 4 times and L repeated twice: $\dfrac{9!}{4!2!} = 7560$ distinct arrangements.
- Three-digit numbers from 0–5, no repetition: first digit has 5 choices (not 0), then 5 and 4 choices: $5 \cdot 5 \cdot 4 = 100$.

Special restrictions: fix required first/last letters or digits first, then arrange the remaining positions. If a leading digit cannot be zero, count the first position separately.

2.3 3. Combinations

A **combination** is a selection where order does not matter: AB and BA are the same selection.

$$\boxed{{}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}}$$

Relationship with permutations:

$$\boxed{{}^nP_r = {}^nC_r\, r!} \qquad \boxed{{}^nC_r = \frac{{}^nP_r}{r!}}$$

Properties to memorise

$$\binom{n}{0} = 1 \qquad \binom{n}{n} = 1 \qquad \binom{n}{1} = n$$

$$\binom{n}{r} = \binom{n}{n-r} \qquad \text{(symmetry)}$$

$$\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r} \qquad \text{(Pascal's identity)}$$

$$\sum_{r=0}^{n} \binom{n}{r} = 2^n$$

Examples

- Choose 3 from 8 students: $\binom{8}{3} = 56$.
- Choose 2 from 6 computers to form a direct link: $\binom{6}{2} = 15$ links.
- Committee of 3 from 5 boys and 4 girls, at least one girl: $\binom{9}{3} - \binom{5}{3} = 84 - 10 = 74$.
- Committee of 4 from 6 men and 5 women, at least one woman: $\binom{11}{4} - \binom{6}{4} = 330 - 15 = 315$.
- Choose 2 black from 5 and 3 red from 6: $\binom{5}{2}\binom{6}{3} = 10 \cdot 20 = 200$. Multiply because each black pair can be combined with each red group.

At least one is often easiest by complement:

$$\boxed{\text{at least one wanted} = \text{all possibilities} - \text{none wanted}}$$

2.4 4. Pigeonhole principle

Basic principle: if $n+1$ objects are placed into n boxes, at least one box contains at least two objects. **Generalised principle:** distributing N objects among k boxes guarantees some box has at least

$$\boxed{\left\lceil \frac{N}{k} \right\rceil \text{ objects}}$$

The ceiling $\lceil x \rceil$ means round up to the next integer. **Guarantee at least r objects in one box:**

$$\boxed{N_{\min} = k(r-1) + 1}$$

Examples

- 13 people and 12 birth months: at least 2 share a month.
- 47 students in 8 groups: $\lceil 47/8 \rceil = 6$, so at least one group has 6 or more.
- Minimum people to guarantee 5 share a birth month: $12(5-1)+1 = 49$.
- 8 students and 7 chairs: if all are seated, at least one chair must be shared (if sharing is allowed).

Exam proof format: identify objects and boxes; assume each box has at most $r-1$ objects; maximum without reaching r is $k(r-1)$; one more object forces a box to contain at least r.

2.5 5. Inclusion–Exclusion principle

Two sets:

$$\boxed{|A \cup B| = |A| + |B| - |A \cap B|}$$

Subtract the overlap once because it was counted twice. Example: 20 like Python, 15 like Java, 5 like both: at least one $= 20 + 15 - 5 = 30$.

Three sets:

$$\boxed{\begin{aligned} |A \cup B \cup C| = {} & |A| + |B| + |C| \\ & - |A \cap B| - |A \cap C| - |B \cap C| \\ & + |A \cap B \cap C| \end{aligned}}$$

Add individual sets, subtract pairwise overlaps, then add the triple overlap back once.

Exactly two sets (three-set case):

$$\boxed{E_2 = |A \cap B| + |A \cap C| + |B \cap C| - 3|A \cap B \cap C|}$$

An element in all three is counted in all three pairwise intersections, so subtract it three times to exclude it from the exactly-two count.

Exactly one set (three-set case):

$$\boxed{\begin{aligned} E_1 = {} & |A| + |B| + |C| \\ & - 2\big(|A \cap B| + |A \cap C| + |B \cap C|\big) \\ & + 3|A \cap B \cap C| \end{aligned}}$$

Worked example: 120 like pizza, 100 burgers, 80 tacos; 60 like pizza and burgers, 30 pizza and tacos, 40 burgers and tacos; 20 like all three.

$$\begin{aligned} \text{At least one} &= 120 + 100 + 80 - 60 - 30 - 40 + 20 = 190 \\ \text{Exactly two} &= 60 + 30 + 40 - 3(20) = 70 \end{aligned}$$

If the total population is N, then **none** $= N - |A \cup B \cup C|$. In the example, if the survey has 200 people, none $= 200 - 190 = 10$.

2.6 6. Mathematical induction

Mathematical induction proves that a statement $P(n)$ is true for every integer n in a specified range, usually $n \geq 1$.

The two essential steps

1. **Base case:** verify $P(1)$ (or the stated smallest value).
2. **Inductive step:** assume $P(k)$ is true for an arbitrary k in the range (the inductive hypothesis), then use that assumption to prove $P(k+1)$.

Once both steps are established, conclude $P(n)$ is true for all integers in the range.

Standard proof template

1. Let $P(n)$ be the statement to prove.
2. *Base case:* For $n = 1$, left side $= \cdots$ and right side $= \cdots$, so $P(1)$ is true.
3. *Inductive hypothesis:* Assume that for $n = k$, $P(k)$ is true. Write the assumed equation clearly.
4. *Inductive step:* For $n = k + 1$, start with the left side. Separate out the final term, substitute the inductive hypothesis, simplify, and obtain the required right side.
5. *Conclusion:* Hence, by mathematical induction, $P(n)$ holds for all integers $n \geq 1$.

Worked example: prove $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$ **for** $n \geq 1$**.**

- Base case $n = 1$: $1 = \frac{1(2)}{2} = 1$. True.
- Assume for $n = k$: $1 + 2 + \cdots + k = \frac{k(k+1)}{2}$.
- For $k + 1$:

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

  This is the required formula with $n = k + 1$.
- Therefore the statement holds for all $n \geq 1$ by induction.

Another common type: divisibility. To prove $3 \mid (n^3 + 2n)$, verify the base case. Then assume $k^3 + 2k = 3m$ for some integer m. For the next case,

$$\begin{aligned} (k+1)^3 + 2(k+1) &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\ &= (k^3 + 2k) + 3(k^2 + k + 1). \end{aligned}$$

Both terms on the last line are divisible by 3, so the next case is divisible by 3 as well.

3 Fast formula picker

Question wording / task	Use
Choose one of several non-overlapping alternatives	Addition rule: add the counts
Complete several successive choices/positions	Multiplication rule: multiply the choices
Arrange all distinct objects	$n!$
Arrange r of n, order matters, no repeats	${}^nP_r = n!/(n-r)!$
Make a length-r sequence from n symbols, repetition allowed	n^r
Arrange objects with identical repeats	$n!/(p_1!p_2!\cdots)$
Select r of n, order irrelevant	$\binom{n}{r}$
At least one of a category	Total minus none (complement)
Guaranteed shared category / minimum guarantee	Pigeonhole: $\lceil N/k \rceil$ or $k(r-1)+1$
Count in either/overlapping sets	Inclusion–Exclusion
Prove a statement for every natural number	Mathematical induction
Check logical formula validity/classification	Truth table or equivalence laws
Translate "all" / "some" statements	Quantifiers $\forall$ / $\exists$
Derive conclusion from premises	Rule of inference

Last-minute checks

- In permutations, order matters; in combinations, it does not.
- A leading digit of a number cannot be zero unless the question explicitly says strings may start with zero.
- For repeated letters, divide by factorials of the repeated-letter counts.
- For "at least one", consider total minus the case with none.
- In three-set inclusion–exclusion, the triple intersection is added back once.
- "Exactly two" subtracts three times the triple intersection; "exactly one" uses the separate formula above.
- In induction, the hypothesis is an assumption for k, not the conclusion. Prove the $k+1$ case using it.
- In implication, $T \to F$ is the only false row.

Prepared as a concise revision guide aligned to the Module 4 and Module 5 lecture topics.